AI MODEL HEALTH MONITORING & AUTOMATED MAINTENANCE SYSTEM

Feature Ticket Document

Draft · 2026-09-29

Priority: P0 must have, P1 should have, P2 nice to have. Size: S (≤1 day), M (2–3 days), L (a week+).

1. Ticket flow



2. Built (for the record)

ID	Feature	Status
FT-01	Multi-model units with isolation	Done
FT-02	Safe model/dataset intake (hash, whitelist, size limit)	Done
FT-03	Adapters: sklearn, XGBoost, LightGBM, CatBoost	Done (last two uncommitted)
FT-04	Task metrics: binary, multiclass, regression	Done
FT-05	5-pillar health + drift with expected baselines	Done
FT-06	Diagnosis + strategy selector	Done
FT-07	Refinement (candidates)	Done
FT-08	7 safety gates	Done
FT-09	Promotion modes + rollback + audit log	Done
FT-10	Alerts (acknowledge / resolve)	Done
FT-11	First-load sandbox (child process)	Done, uncommitted
FT-12	Fleet UI, guided demo, inference sandbox	Done

3. Open tickets

ID	Title	Pri	Size	Acceptance criteria
FT-13	Review and commit uncommitted work	P0	S	All tests pass; docs updated; commits split by topic; `IntakeView` removal checked
FT-14	Users and roles	P0	L	Login; roles Viewer/Engineer/Approver/Admin; 403 for blocked actions; tests per role
FT-15	Real actor in audit log	P0	S	Every audit row holds the real user name, not "USER"
FT-16	Frontend sends auth token	P0	S	UI works with `MODEL_HEALTH_API_KEY` set; 401 shows a clear message
FT-17	Pytest suite + test database	P1	M	Scripts moved to `pytest`; tests never touch the dev DB; one command runs all
FT-18	Lock down demo routes	P1	S	`/api/demo/*` disabled unless `DEMO_MODE=1`
FT-19	Background job queue for refinement	P1	L	Refine returns a job id at once; UI polls; API not blocked
FT-20	Shared rate limiter	P2	S	Limits hold across several server processes
FT-21	Container sandbox for model load	P2	L	First load runs with no network and low rights; malicious test pickle cannot write files
FT-22	Split large React components	P2	M	No component above ~500 lines; no behaviour change
FT-23	Frontend tests	P2	M	Tests for upload, promote, rollback flows
FT-24	Research CSV export in UI	P2	S	Button downloads output of `export_research_dataset.py`
FT-25	Full ranking metrics (NDCG, MAP)	P2	M	Extends `tasks/ranking.py` beyond Precision@K
FT-26	Trim `ARCHITECTURE.md`	P2	S	Matches code; links to these short docs

4. Ticket template

```
ID / Title:
Why (one sentence):
What changes:
Acceptance criteria (testable):
Out of scope:
Depends on:
```